Semester…3rd..
(Regular&Back)(Back)
Sub & Code…WT (IT2002)..
Branch (s):CSE, IT, CSSE, CSCE.

AUTUMN MID SEMESTER EXAMINATION-20

School of Computer Engineering

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

DEEMED TO BE UNIVERSITY, BHUBANESWAR-24.

WEB TECHNOLOGY

[IT 2002]

Full Marks: 20 **Time: 1.5 Hours**

Answer any FOUR questions including question No.1 which is compulsory.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

1. **(a)**
```
class selection_statements{
 public static void main(String args[]){
int var1 = 5;
int var2 = 6;
if ((var2 = 1) == var1)
System.out.print(var2);
else
System.out.print(++var2);
}
}
```
What is the output of the code snippet given above?

(b) Is it compulsory for an abstract class to have at least one abstract method? Can an abstract method be declared private? Justify both answers.

(c)
```
class multidimension_array
    {
        public static void main(String args[])
        {
            int arr[][] = new int[3][];
            arr[0] = new int[1];
            arr[1] = new int[2];
            arr[2] = new int[3];
          int sum = 0;
          for (int i = 0; i < 3; ++i)
              for (int j = 0; j < i + 1; ++j)
                    arr[i][j] = j + 1;
          for (int i = 0; i < 3; ++i)
              for (int j = 0; j < i + 1; ++j)
                    sum + = arr[i][j];
          System.out.print(sum);
        }
    }
```

What is the output of the code snippet given above? Justify.

(d) What is the HTML syntax for inserting a © symbol in a browser page?

(e) Write down the HTML syntax for displaying ordered and unordered lists on a web page.

[1x5]

2. (a) Create an html form which will have the following fields:

Name	-textbox
Address	-textarea
Password	-password field
Sex	-two separate radio button for male and female
Dob	-select option (three separate option for Date, month and year)
Hobbies	-checkbox (minimum three hobbies)
State	-textbox
Submit Button	-button

[3]

(b) Design the following table using colspan and rowspan:

<table>
<tr><td colspan="3">A</td></tr>
<tr><td rowspan="2">B</td><td colspan="2">C</td></tr>
<tr><td>D</td><td>E</td></tr>
</table>

[2]

3. (a) Write a program that displays the results of a series of 10 coin tosses. Use Math.random() function to generate a number between 0 and 1. After each coin toss, display whether the toss represents ‘heads’ or ‘tails’. If the result is 0.5 or less, the result represents ‘heads’; otherwise, it represents ‘tails’. **[3]**

(b) Write a program in Java to implement the concept of dynamic method dispatch using method overriding. **[2]**

4. (a) Write a program in Java to implement constructor overloading. **[2.5]**

(b) Write a program that prompts the user to input two numbers, *first* and *diff*. The program then creates a one dimensional array of 10 elements and initializes them with an arithmetic sequence. The first number of the sequence is the *first* value and the next number is generated by adding the *diff* to the number preceding it. This formula is repeated for the rest of the sequence. E.g. if *first* =11 and *diff* = 4, then the arithmetic sequence will be 11, 15, 19, 23, 27, 31 … and so on.

[2.5]

5. Write short notes on (any two):

(a) HTML style sheets **(b)** Frames

(c) This and super keywords **(d)** JVM, JDK, JRE **[2.5+2.5]**